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C:\Windows\System32\cmd.exe - mysql -u root -p
Microsoft Windows [Version 10.0.22000.2176]
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C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database libflow;
Query OK, 1 row affected (0.01 sec)

mysql> use libflow;
Database changed
mysql> CREATE TABLE Books (
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100) NOT NULL,
  -> isbn VARCHAR(20) UNIQUE,
  -> published_year INT CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.10 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn                | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565      | 1925          |
| 2       | To Kill a Mockingbird | 9780061120084      | 1960          |
| 3       | 1984                 | 9780451524935      | 1949          |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,

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-> full_name VARCHAR(100),
-> email VARCHAR(100) UNIQUE
-> );
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Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Members (member_id, full_name, email) VALUES

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-> (101, 'John Smith', 'john.smith@email.com'),
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),
-> (103, 'Michael Brown', 'michael.brown@email.com');
```

Query OK, 3 rows affected (0.02 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> CREATE TABLE Loans (

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-> loan_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> loan_date DATE,
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.11 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES

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-> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

Query OK, 10 rows affected (0.02 sec)

Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Loans; SELECT m.full_name AS Member_Name, b.title AS Book_Title

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10 rows in set (0.00 sec)

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-> FROM Loans l
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-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;SELECT published_year, COUNT(book_id) AS Total_Books
+-----+
| Member_Name | Book_Title |
+-----+
| John Smith  | The Great Gatsby |
| John Smith  | To Kill a Mockingbird |
| John Smith  | 1984 |
| John Smith  | The Great Gatsby |
| Emma Wilson | To Kill a Mockingbird |
| Emma Wilson | The Great Gatsby |
| Emma Wilson | 1984 |
| Michael Brown | 1984 |
| Michael Brown | To Kill a Mockingbird |
| Michael Brown | The Great Gatsby |
+-----+
10 rows in set (0.01 sec)

-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;CREATE TABLE Donation_History (
+-----+
| published_year | Total_Books |
+-----+
| 1925 | 1 |
| 1949 | 1 |
| 1960 | 1 |
+-----+
3 rows in set (0.00 sec)

-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.50 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.01 sec)

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Query OK, 1 row affected (0.01 sec)

mysql> COMMIT;CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.01 sec)

Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780451524935';

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